National University of Computer and Emerging Sciences

Software Re-Engineering (SE4001)

Sessional-II Exam

Date: April 9, 2026

Course Instructor(s)

Dr. Farooq Ahmed, Mr. Arslan Asif

Total Time: 1 hour

Total Marks: 30

Total Questions: 2

______ Roll No ______ Section ______ Student Signature

Do not write below this line

Attempt all questions on the answer sheet

CLO 3: Apply different approaches to understand code

Question 1 **[10+5 marks]**

(a) Consider a requirement to parse Javascript function calls and identify the number of parameters in each function. Following are a few code snippets to illustrate function calls, assuming that definitions are already made:

`print()`	`sort(array)`	`search(array,key)`	`distance(1,1,3,4)`

Write BNF rules to specify function call syntax, so that parsing can be done.

(b) Show a derivation / parse tree for **`distance(1,1,3,4)`** call

CLO 5: Implement refactoring strategies for effective re-engineering

Question 2 **[5+10 marks]**

Consider the code of LibraryMember class (provided on next page) taken from a Library Management application. Code review reveals some code smells / design issues with the class that can be addressed through multiple refactorings to make the design more extensible and maintainable.

(a) Which refactoring techniques will be most appropriate in this context? List as many as applicable along with a one-line motivation for each.

(b) Write final refactored code after applying relevant refactorings.

```
class LibraryMember {
  String _name;
  int _daysLate;
  String _membershipType; // "REGULAR", "STUDENT", "PREMIUM"
  int _booksBorrowed;

  public LibraryMember(String name, int daysLate, String membershipType,
                       int booksBorrowed) {
    this._name = name;
    this._daysLate = daysLate;
    this._membershipType = membershipType;
    this._booksBorrowed = booksBorrowed;
  }

  public void processMember() {

   // Temp variables
   double fine = 0;
   double discount = 0;

   // Fine calculation
   if (_daysLate > 0) {
     fine = _daysLate * 10;
     if (_membershipType.equals("STUDENT")) {
       fine = fine * 0.5;
     }
     else if (_membershipType.equals("PREMIUM")) {
       fine = fine * 0.2;
     }
   }

   // Discount logic
   if (_booksBorrowed > 5) {
     if (_membershipType.equals("PREMIUM")) {
       discount = 50;
     }
     else if (_membershipType.equals("STUDENT")) {
       discount = 20;
     }
   }

   double total = fine - discount;

   // Displaying member info and due amount details
     System.out.println("Member: " + _name);
     System.out.println("Membership: " + _membershipType);
     System.out.println("Days Late: " + _daysLate);
     System.out.println("Books Borrowed: " + _booksBorrowed);
     System.out.println("Fine: " + fine);
     System.out.println("Discount: " + discount);
     System.out.println("Total Due: " + total);
  }
}
```

```
public class LibraryApp {
  public static void main(String[] args) {
    LibraryMember member = new LibraryMember("Sara", 7, "STUDENT", 6);
    member.processMember();
  }
}
```